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X-ray Investigations of Sol–Gel-Derived GeO₂ Nanoparticles

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Abstract: Germanium dioxide (GeO₂) is a versatile material with several different crystalline polymorphs and interesting applications in, e.g., optics, microelectronics, and Li-ion batteries. In particular, many of the material's properties depend on the size of the prepared crystallites, and thus, nanocrystalline GeO₂ is of special interest. Here, GeO₂ nanoparticles are prepared via sol–gel processes by the hydrolysis of Ge isopropoxide (Ge(OCH(CH₃)₂)₄). The precipitated powders are dried at room temperature and annealed in ambient air using temperatures between 500 °C and 1000 °C from 3 to 24 h. The samples were characterized by X-ray diffraction, X-ray absorption fine structure spectroscopy, and scanning electron microscopy, providing the crystalline structures, the phase composition, as well as the morphology and crystallite size of the formed particles and their changes upon heating. According to the structural analysis, the samples are crystalline with a dominant β- (low temperature) quartz phase without any heat treatment directly after drying and increasing contributions of α- (high-temperature modification) quartz and quartz-like GeO₂ structures with increasing temperature and annealing time were found. According to electron microscopy and the X-ray analysis, the particle size ranges from about 40 to 50 nm for the pristine particles and to about 100 nm and more for the annealed materials.

Keywords: germanium dioxide; X-ray absorption spectroscopy; X-ray diffraction; solution chemistry; crystallite size



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1. Introduction

Oxide nanoparticles play an important role in various areas of physics, chemistry, and materials science [1], with numerous applications in microelectronic circuits [2], catalysts [3,4], sensors [5–7], Li-ion batteries [8,9], and pharmaceuticals [10], to mention just a few. Many properties of the particles are governed by their particle size. For example, the importance of the surface free energy increases with decreasing particle size, which may lead to changes in the unit cell parameters or to more complex structural phase transitions in the case of materials such as Al₂O₃, Fe₂O₃, TiO₂, and other binary, ternary, or quaternary oxides [11–14]. This is especially true for oxide systems, in which different stable polymorphs exist, such as, e.g., TiO₂, where the anatase phase is thermodynamically stable for a crystallite size below ca. 11 nm, brookite is stable in an intermediate range from 11 nm to ca. 35 nm, while the rutile phase is favorable for crystallites larger than 35 nm in size [15].

In this particular context, GeO₂ is a very interesting oxide material from a fundamental point of view because the polymorphism of GeO₂ is very complex, revealing different Ge–O coordination environments as follows. In addition to the tetragonal argutite structure, which is identical to the rutile TiO₂ structure with lattice constants of 4.396 Å and 2.863 Å and a Ge–O coordination number of 6 [16], several different hexagonal forms exist, i.e., the α-quartz-structure, also called high-temperature quartz [17] and the β-quartz structure (low-temperature quartz, JCPDF 36-1463 [18]). Furthermore, other modifications have been produced under specific conditions [19,20], in particular, the so-called quartz-like structure [21], resulting in a total of four different crystalline GeO₂ polymorphs. In the rutile-type GeO₂ phase, Ge is coordinated with the six nearest oxygen, while in the three

mentioned quartz structures, Ge is only 4-coordinated with oxygen. The quartz and quartz-like structures are very similar, i.e., α - and β -quartz both crystallize in space-group 152, with only minor differences in the unit cell parameters and slightly different positions of Ge and O within the unit cell and accordingly slightly different Ge-O and Ge-Ge distances and bond angles. While β -quartz is assumed to be stable below a temperature of $T = 573\text{ }^{\circ}\text{C}$, α -quartz is supposed to dominate above this temperature, respectively. Accordingly, β -quartz is also named the low-temperature modification [18], while α -quartz is the high-temperature phase [22,23]. The quartz-like structure has space-group 154, which is quite similar to 152, and the atom positions in the unit cell are similar to those of the other two structures, again with slightly different bond distances [21]. Structural motives and the unit cell parameters of all involved GeO_2 structures are compiled in the Electronic Supplemental Material, Figures S1–S4. Furthermore, calculated diffraction patterns are also provided in Figure S5 in the supplement. From this figure and the literature, it can be concluded that the diffractograms of the three quartz structures are all very similar, with only slightly different Bragg peak positions, however, with larger intensity differences. Nevertheless, the three hexagonal GeO_2 quartz structures can be well distinguished from the rutile GeO_2 modification. Last but not least, also amorphous GeO_2 (GeO_2 glass) is observed [24]. Thus, it appears interesting to study stable GeO_2 structures in the case of nanoparticles also from a fundamental point of view.

Second, there are different potential applications for GeO_2 nanoparticles. Compared to SiO_2 , GeO_2 has a substantially larger dielectric constant, which makes GeO_2 nanoparticles interesting for applications in micro- and nano-electronics, e.g., as a capacitor dielectric [25], and it also has a larger refractive index that can be useful for optoelectronic applications [26]. Furthermore, GeO_2 in the form of nanowires obeys a tuneable optical luminescence [27]. Very recently, GeO_2 nanoparticles were used as anode material in a Li-ion battery with very promising properties in terms of charge capacity and capacity fading [28,29]. Last but not least, GeO_2 nanoparticles can be used as precursors for the synthesis of other Ge-containing nanomaterials, such as, e.g., Ge sulphides [30].

For the GeO_2 nanoparticle preparation, several different routes have been described in the literature in the past. In many cases, sol–gel processes are used to synthesize nanostructures, in particular for catalytic applications. The opportunity to easily prepare large amounts of oxide material, which additionally have a large surface area, is an advantage of sol–gel processes. Furthermore, different possibilities are known to influence the reaction products of this wet-chemical preparation route, e.g., by performing the reaction in alcoholic or acidic environments, using elevated temperatures and pressures, or by adding additional anions or cations in the solutions (see, e.g., [31,32]). While the preparation routes for SiO_2 employing, e.g., dilute Si ethoxides and appropriate additives for the synthesis of Si oxides are well-developed, however, using Ge ethoxide as a precursor for the Ge-oxide preparation, the fast hydrolysis rate makes any control of the reaction and thereby also the tailoring of the final GeO_2 nanoparticle morphology difficult. Thus, the amount of water used in a reaction involving the Ge-ethoxide species must be minimized and carefully controlled. Therefore, microemulsion techniques using capping agents have been developed, leading to hexabranched GeO_2 particles with a size of about 180 nm and a β -quartz structure without any additional treatment [29,33]. Other groups have applied biomineralization techniques using both Ge ethoxide as well as Ge isopropoxide as precursors; however, exhibiting amorphous structures directly after preparation only [26]. Heat treatment of sol–gel-derived GeO_2 particle films showed that crystallization into a hexagonal phase starts at temperatures of ca. $400\text{ }^{\circ}\text{C}$ [25], while prolonged heating above $650\text{ }^{\circ}\text{C}$ was required to crystallize amorphous precipitates obtained from the hydrolysis of Ge isopropoxide [32], similar, e.g., to titania nanoparticles [13,15].

In the present work, GeO_2 particles were prepared by the hydrolysis of Ge isopropoxide in pure water without using capping agents or acidic catalysts. The prepared nanoparticles were investigated by a combination of X-ray diffraction (XRD), X-ray absorption fine structure spectroscopy (XAFS), and scanning electron microscopy (SEM). While XRD

provides quantitative information about the crystalline phase composition and the particle size, XAFS may provide the atomic short-range order structure independent from the structural regularity of the material. Due to the similarity of the involved GeO₂ phases, X-ray diffraction alone may not be sufficient to resolve the small structural differences of the prepared samples, and thus, complimentary and supporting X-ray absorption spectroscopy may provide valuable information and support the conclusions from a detailed analysis of the X-ray diffraction patterns.

SEM, in addition, provides the morphology of the particles and also gives access to the particle size and shape. The aim of this work is, thus, to investigate the development of the different GeO₂ phases in the fresh precipitates as well as after heat treatments for different temperatures and heating times.

2. Materials and Methods

2.1. GeO₂ Synthesis Procedures

The germanium oxide powders are prepared via room temperature hydrolysis and condensation of pure Ge isopropoxide (Ge[OCH(CH₃)₂]₄) (from Sigma Aldrich, Darmstadt, Germany with a purity of 97%) in de-ionized water. A total of 4 mL of Ge isopropoxide in 50 mL of water were used. For comparison, an additional sample was prepared in drinking water to assess the influence of anions and cations on the structure formation. The white precipitates were filtered, dried at room temperature for at least 8 h, and pestled in a mortar prior to the structural investigations. All annealing treatments were carried out in an industrial furnace in ambient air at 500 °C (3 h), 800 °C (3 h, 5 1/2 h, 24 h), and 1000 °C (3 h, 24 h), respectively. Those temperatures were chosen according to the literature since any substantial structural changes of the GeO₂ deposits are not observed for annealing temperatures below ca. 500 °C [25,32,34,35]. A heating ramp with 10 °C/min was applied to reach the final annealing temperature. After cooling down overnight in the furnace, the samples were stored in air. Up to now, eight different samples and heat treatments have been performed.

2.2. X-ray Diffraction (XRD) Experiments

For the XRD measurements of the as-prepared and annealed Ge-oxide powders, an X'Pert Pro multi-purpose diffractometer (PanAnalytical, Almelo, the Netherlands) was employed using non-monochromatized Cu K α radiation in the Bragg–Brentano geometry and a multi-strip Si detector (X'celerator). Applying an acceleration voltage of 40 kV and electron emission currents of 40 mA, the diffraction patterns were collected for Bragg angles 2 θ between 10° and 150°. The powder samples were dispersed on a rotating holder for the measurements to improve the counting statistics.

Phase identification was done using the Bragg peak positions and intensities related to the different crystalline GeO₂ phases in question, as detailed above. The particle sizes of the GeO₂ nanocrystallites were determined from the diffraction patterns using the Scherrer formula.

$$d[\text{nm}] = \frac{k \cdot \lambda[\text{nm}]}{(B^2(2\theta) - B_0^2(2\theta))^{1/2} \cdot \cos(\theta)}$$

With the X-ray wavelength λ , the peak broadening $B(2\theta)$ of a Bragg reflection detected at an incident angle θ , and assuming that the Bragg peaks have a Gaussian shape. $B_0(2\theta)$ represents the instrumental peak broadening; it was experimentally determined from diffraction experiments of different single crystals. The parameter k accounts for the particle's shape, which was set to $k = 0.9$ here, according to the approximately spherical or cubic shape of the detected crystallites using scanning electron microscopy (see below). Furthermore, the particle size was also calculated using the Williamson–Hall method [36], which is implemented in the PowderCell software 2.4 [37].

2.3. X-ray Absorption Fine Structure (XAFS) Experiments

Transmission mode XAFS experiments were carried out at beamline 10 [38] at the DELTA storage ring (Dortmund, Germany), operating with 1.5 GeV electrons with ~100–150 mA of stored current. A Si(111) channel-cut monochromator was used for the XAFS studies at the Ge K-edge at 11,104 eV. Incident and transmitted intensities were measured using nitrogen- and argon-filled ionization chambers and a Ge-reference sample was fixed between the second and a third ionization chamber in order to properly calibrate the energy scale of the monochromator simultaneously with each sample. The samples for the XAFS measurements were prepared by dispersing the GeO₂ powders on self-adhesive tape and stacking several tapes in order to obtain an absorption suited for the transmission mode experiments.

XAFS data reduction comprised the spectral calibration, pre-edge as well as post-edge background removal, edge determination, normalization, and fitting of the extracted XAFS-function $\chi(k)$ and was done using the Demeter software package [39]. For comparison, a β -quartz GeO₂ reference sample (99.999% purity from Sigma Aldrich), a Ge(111) single crystal, amorphous GeO (ref. [40]), as well as the Ge isopropoxide (Ge[OCH(CH₃)₂]₄) precursor were also measured.

2.4. Scanning Electron Microscopy

SEM experiments were conducted in a JEOL JSM 6510 scanning electron microscope equipped with an energy-dispersive X-ray (EDX) analysis system (Noran 7, Thermo Fisher, Madison, WI, USA). The powder samples were dispersed on conductive, self-adhesive tapes without any pre-treatment and investigated under high-vacuum conditions. EDX spectra were measured for several particles of each prepared sample, and the numbers were averaged in order to obtain statistical relevance.

3. Results

3.1. Scanning Electron Microscopy

In Figure 1, some exemplary SEM micrographs of prepared GeO₂ nanoparticle samples are presented. A large number of agglomerated nanoparticles with a size of about 100 nm and larger are visible, independent of the heat treatment. Many of the particles have a cubic shape; however, some of the visible particles seem to reveal circular shapes as well, which can, however, not be fully resolved with the used electron microscope. Those observations qualitatively meet previous observations; however, the crystallite size observed in the present study appears to be significantly smaller [30]. In order to elucidate the size of the particles more precisely, the measured SEM micrographs were subjected to a power spectral density (PSD) analysis, see, e.g., [41–43]. Here, if objects of similar size are present in the sample, peaks at a characteristic frequency will appear in the power spectrum, and the size of these features can be determined accordingly [41–43]. In contrast, if a randomly rough surface is inspected, a plateau will appear in the spectrum at low frequencies [41]. We have analyzed the SEM micrographs using the Gwyddion software package (Version 2.63) [44], and the PSD functions determined from the SEM micrographs in Figure 1a–c are displayed in Figure 1d. As can be seen, prominent peaks appear in the frequency domain well below 0.02 nm^{-1} , which can be related to the size of the investigated GeO₂ nanoparticles. For the inspected images, the peak frequency relates to an average particle size of $125.3 \text{ nm} \pm 34.4 \text{ nm}$ for the pristine sample prepared in de-ionized water, $125.3 \text{ nm} \pm 34.4 \text{ nm}$ for the sample annealed at 800 °C for 3 h, and $125.3 \text{ nm} \pm 34.4 \text{ nm}$ for the sample annealed at 1000 °C for 24 h. As can be seen, a slight increase in the average particle size is observed.

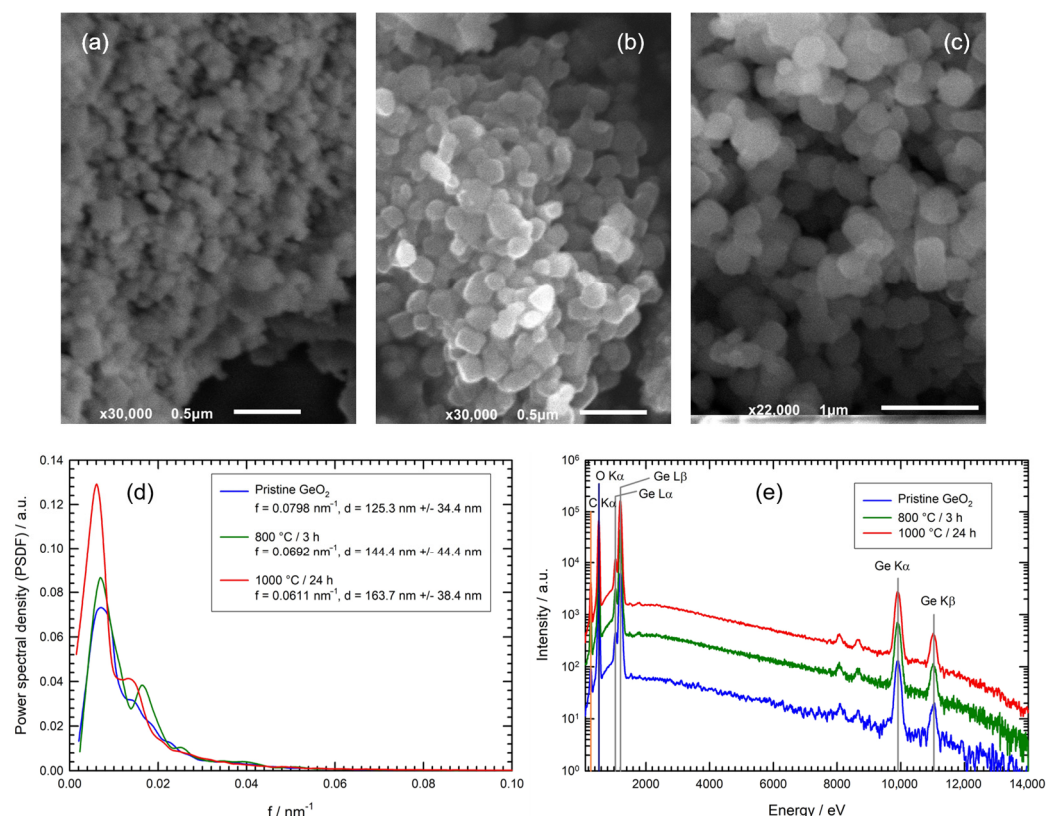


Figure 1. Selected examples for SEM investigations of GeO₂ particles prepared by hydrolysis after drying at room temperature. (a) Pristine sample directly after drying and after heating for 3 h at 800 °C (b) and at 1000 °C for 24 h (c). (d) Power spectral density (PSD) for the three images (a–c) and the determined average lateral size of the crystallites. (e) EDX analysis of the three samples, with elemental contributions of C, O, and Ge as indicated. The spectra are vertically shifted for clarity. The elemental compositions are $66.95 \pm 0.59\%$ O, $30.37 \pm 1.61\%$ Ge, and $2.68 \pm 0.40\%$ C for the pristine sample, $63.38 \pm 0.64\%$ O, $33.81 \pm 0.21\%$ Ge, and $2.81 \pm 0.44\%$ C for the sample heated at 800 °C for 3 h, and $65.86 \pm 0.34\%$ O, $28.04 \pm 0.45\%$ Ge, and $5.67 \pm 0.12\%$ C for the sample heated at 1000 °C for 24 h, respectively. At 8000 eV and 8900 eV, small Cu peaks from the sample holder are visible, which corresponds to an elemental contribution of ca. 0.4% and less.

According to the EDX analysis, see Figure 1e, the elemental composition of the prepared materials is very close to the targeted Ge/O ratio of 1:2, i.e., a quantitative analysis provided average concentrations of $64.1 \pm 2.6 \text{ at.}\%$ O, $33.2 \pm 2.5 \text{ at.}\%$ Ge, and only $2.6 \pm 0.2 \text{ at.}\%$ C. Only spurious signals related to the Cu sample holder smaller than ca. 0.4% were detected, but no other contaminations were detectable, in good agreement with previous studies employing sol–gel processes [25]. These values show that the Ge-oxide material contains only a very small amount of carbon contaminations, possibly due to the contact with the laboratory air and the vacuum system of the SEM, or remainders from the Ge isopropoxide precursor material, as well as from the conductive tape used to mount the samples in the SEM. The determined Ge/O ratio has a value of $1/1.9 \pm 0.1$, which implies the absence of excessive water from the hydrolysis and almost stoichiometric GeO₂ particles.

3.2. XRD Measurements

Diffraction patterns obtained from all the prepared GeO₂ samples are presented in Figure 2. By comparison with the calculated diffraction patterns (see Figure S5 of the supplement), small but subtle differences between the different samples may be recognized, in particular, in the zoomed view in Figure 2b, different peak intensities, and small peak shifts, as well as different peak splittings, are observed. It is interesting to note that well-

developed diffraction peaks are already observed for the pristine sample after drying at room temperature, without any additional heat treatment, in contrast to what has been published in the literature (see, e.g., [34,45]). However, crystalline deposits were recently also observed for a room-synthesis of GeO_2 , with a dominating β - GeO_2 structure, however, with a predominant spherical shape of the precipitates [46]. These authors proposed a fast Ostwald ripening to be responsible for the crystalline structures, even at room temperature [46]. For the present investigation, the formation of nanosized crystalline particles is in agreement with the fast reaction speed observed during the Ge-isopropoxide hydrolysis, which may facilitate the rapid formation of numerous few-atom GeO_2 nuclei that prevents, on the one hand, the growth of larger crystallites, but, due to the use of a concentrated precursor solution, may lead to aggregated crystalline nanoparticles in the course of the hydrolysis reaction. Furthermore, the less rigid network of GeO_2 in comparison to SiO_2 makes solid-state crystallization possible without heat treatments.

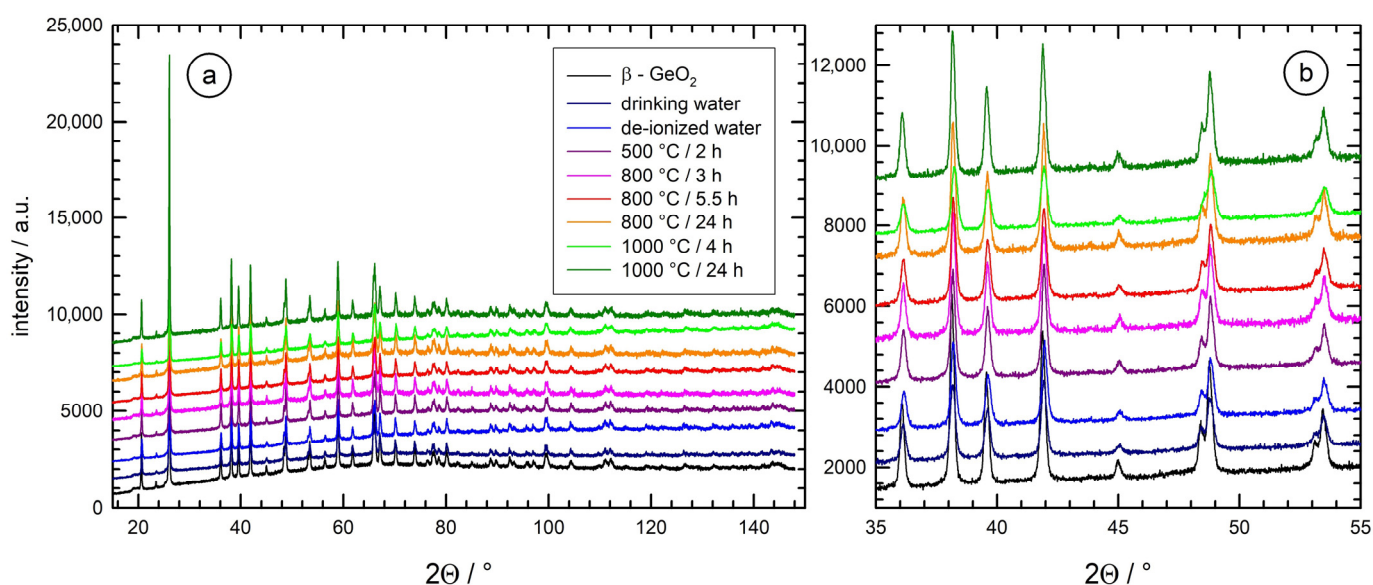


Figure 2. (a) X-ray diffraction patterns from a β - GeO_2 reference sample and GeO_2 precipitates obtained after hydrolysis of Ge isopropoxide at room temperature and different annealing procedures as indicated. In (b), a zoomed view is presented. The experimental data are vertically shifted with respect to each other to facilitate the discrimination of the individual curves.

Fits of the diffraction data of GeO_2 samples in the as-prepared state employing de-ionized water without any subsequent heat treatment and after annealing at $800\text{ }^\circ\text{C}$ for 5.5 h are depicted in Figure 2a,b, respectively. Many diffraction peaks are detectable in the investigated angular range already without any heat treatment, and for comparison, the line positions of α - GeO_2 , β - GeO_2 , the quartz-like, and the rutile phase are indicated in the lower panels of the graphs. Most obvious is a substantial intensity increase of the observed diffraction peaks due to the heating, but also small peak shifts of the diffraction peaks as well as peak shape changes are observed, as can be seen in Figure 2b and the inserts in Figure 3. The experimental data can be well fitted using a Rietveld refinement employing the PowderCell software (version 2.4) [37], making use of the four above-mentioned crystalline GeO_2 structures as input parameters for the fits in the entire angular range measured. Purely Gaussian peak profiles were used, employing the Caglioti formalism for the peak broadening [47] implemented in the PowderCell software, and thus, the width of the diffraction peaks was not refined as an independent parameter. The results of the refinements are compiled in Figure 4 and Table 1, where the determined phase fractions for all the investigated samples are displayed.

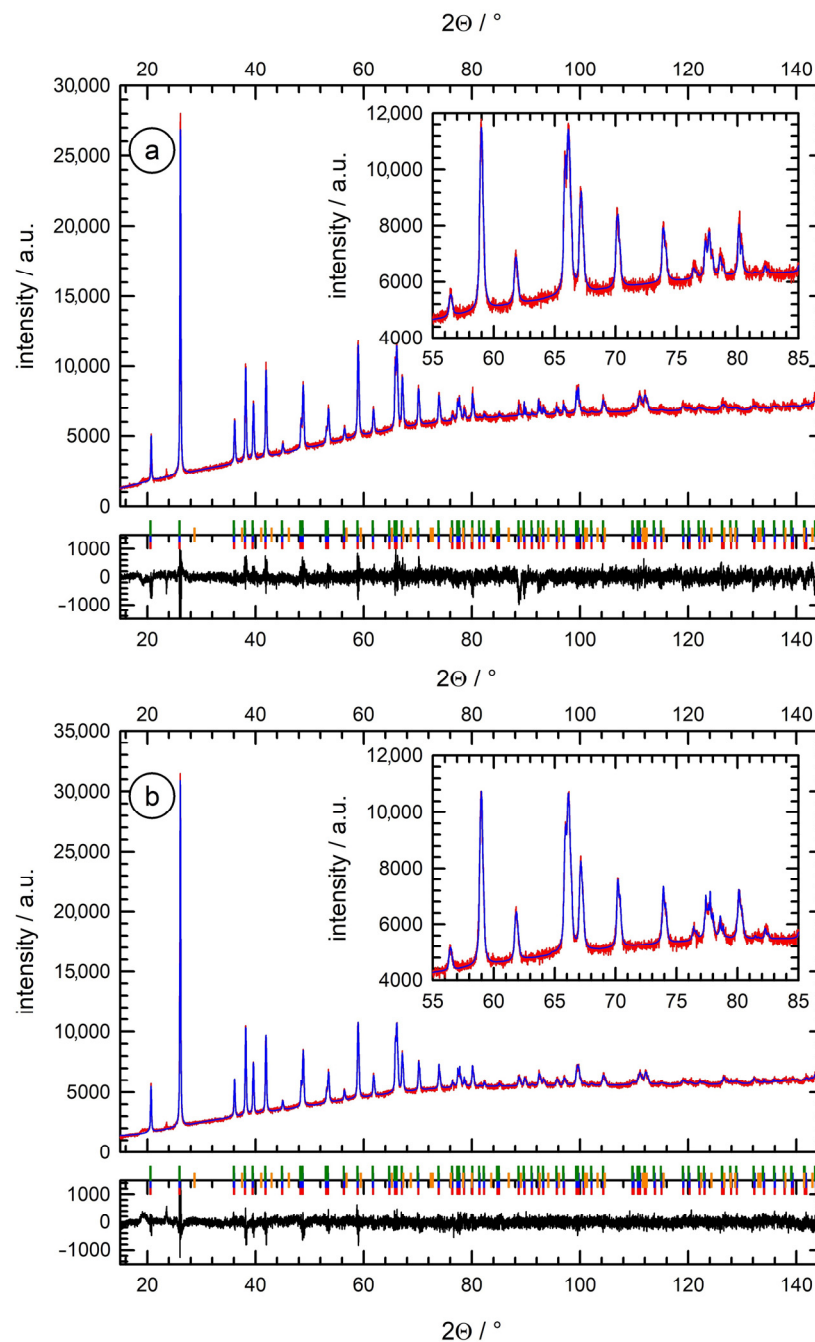


Figure 3. X-ray diffraction patterns from a pristine GeO_2 sample prepared in de-ionized water (a) and a sample annealed in air at $800\text{ }^\circ\text{C}$ for 5.5 h (b). The experimental data (red lines) are fitted using the Rietveld refinement implemented in PowderCell [37]. The inserts depict enlarged views of the range between 55° and 85° in the Bragg angle, indicating that also smaller Bragg peaks are well modeled by the fit. The lower panels of the graphs show the fit residuals. Quantitative results obtained $88.7 \pm 2.4\%$ of the low-temperature β -quartz modification and $11.3 \pm 0.3\%$ of the quartz-like form for the pristine GeO_2 sample with an R -factor of 0.007, and $61.5 \pm 1.6\%$ of α -quartz, $10.1 \pm 0.3\%$ of β quartz, and $28.4 \pm 0.7\%$ of the quartz-like structure for the annealed GeO_2 sample ($R = 0.004$), respectively. Peak positions for the different GeO_2 modifications are indicated by colored vertical bars, i.e., α -quartz (red), β -quartz (green), quartz-like GeO_2 (blue), and rutile-type GeO_2 (orange), respectively.

Table 1. Compilation of lattice parameters and the phase fractions calculated from the fits of the XRD and the EXAFS data, respectively. See text for more details on the fits. In the heat treatments marked with an asterisk (*), the fit results provided very small concentrations for the β -GeO₂ phase only; thus, the obtained lattice parameters are expected to be afflicted with a large error and are thus provided in brackets only.

Sample		XRD—Results			EXAFS Fit Results		
		Conc.	a/Å	c/Å	Conc.	a/Å	c/Å
β -GeO ₂ reference	β -quartz	100%	4.988 (4)	5.647 (2)	93.1%	4.9921	5.6578
	Quartz-like	--	--	--	6.9%	4.9971	5.6635
	α -quartz	--	--	--	--	--	--
Drinking water	β -quartz	97.0%	4.991 (4)	5.653 (9)	74.5%	4.9930	5.6588
	Quartz-like	3.0%	4.9573	5.6951	25.5%	4.9814	5.6457
	α -quartz	--	--	--	--	--	--
De-ionized water	β -quartz	88.7%	4.992 (3)	5.655 (1)	71.7%	4.9894	5.6547
	Quartz-like	11.3%	4.9809	5.6494	28.3%	4.9884	5.6536
	α -quartz	--	--	--	--	--	--
500 °C 2 h	β -quartz	33.9%	4.987 (5)	5.650 (7)	49.9%	5.0018	5.6687
	Quartz-like	6.6%	(5.0871)	(5.7360)	30.4%	4.9771	5.6408
	α -quartz	59.5%	4.9868	5.6487	19.6%	4.9314	5.6034
800 °C 3 h	β -quartz	3.9%	(5.0818) *	(5.5443) *	47.1%	5.0037	5.6710
	Quartz-like	50.0%	4.9889	5.6452	23.8%	4.9673	5.6297
	α -quartz	46.1%	4.9881	5.6490	29.1%	4.9265	5.5978
800 °C 5 ½ h	β -quartz	10.1%	4.996 (1)	5.587 (8)	37.9%	5.0072	5.6749
	Quartz-like	28.4%	4.9901	5.6463	10.2%	4.9485	5.6084
	α -quartz	61.5%	4.9902	5.6488	51.9%	5.0014	5.6830
800 °C 24 h	β -quartz	1.7%	(5.0818) *	(5.5444) *	--	--	--
	Quartz-like	48.6%	4.9893	5.6470	--	--	--
	α -quartz	49.8%	4.9875	5.6491	--	--	--
1000 °C 4 h	β -quartz	--	--	--	11.0%	4.9531	5.6136
	Quartz-like	39.0%	4.9950	5.6453	30.5%	4.9355	5.5936
	α -quartz	61.0%	4.9889	5.6491	58.5%	5.0050	5.6870
1000 °C 24 h	β -quartz	12.0%	5.002 (2)	5.544 (3)	8.0%	4.9976	5.6640
	Quartz-like	29.0%	4.993 (3)	5.643 (1)	28.0%	4.9630	5.6248
	α -quartz	59.0%	4.990 (4)	5.646 (9)	64.0%	5.0157	5.6992

First of all, it has to be noted that argutite, the rutile type GeO₂ modification, could not be proved in any of the investigated samples, implying that no evidence for 6-fold coordinated Ge could be given. The quantitative evaluation shows that the pristine samples are mainly composed of the low-temperature β -GeO₂ phase with a content of typically 90%, which progressively decreases with annealing temperature and annealing time, i.e., after 2 h of annealing at 500 °C, only $\approx$ 34% of β -GeO₂ was determined, and for 3 h at 800 °C, this value decreases to less than 5%. The obtained results suggest that the decrease in β -GeO₂ concentration occurs due to the increase of the high-temperature α -GeO₂ form and the quartz-like structure. Due to the similarity of the latter two phases, their exact quantities are afflicted with uncertainties of typically $\pm$ 2%; however, their generally increasing trends are obvious from the fits of the XRD data.

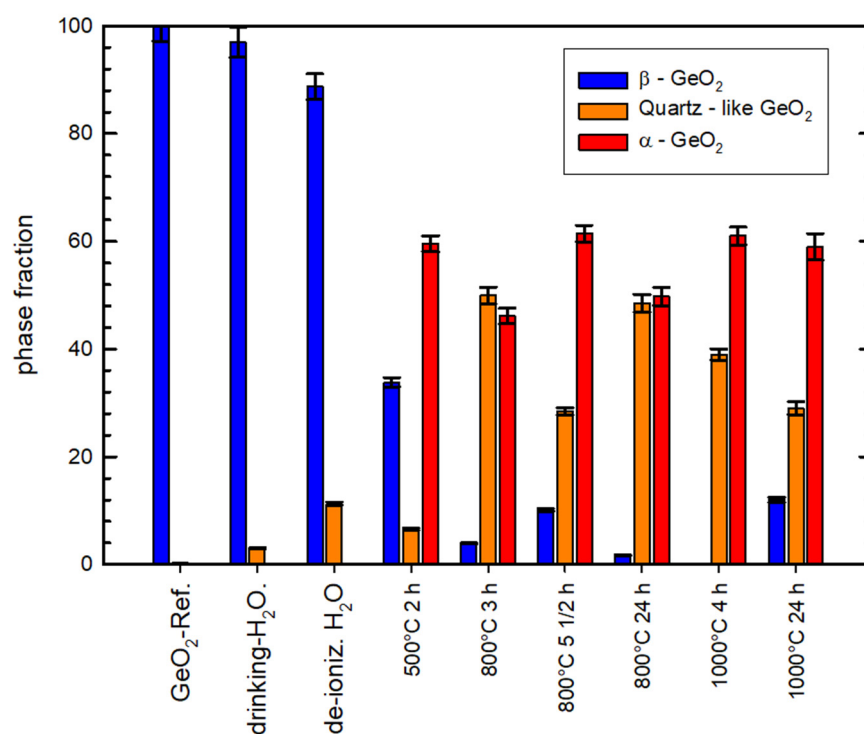


Figure 4. Phase fractions determined by a quantitative analysis of the XRD data. Error bars indicate the confidence interval of the fits. The rutile-type argutite structure could not be proven in any of the samples.

The sizes of the diffracting domains were calculated based on the Scherrer formula and Williamson–Hall (W–H) plots, and the obtained results are compiled in Figure 5. Some representative results for the conducted W–H plots are depicted in Figure S6 of the Supplementary Material. As can be seen, both techniques provided virtually the same values for each sample, ranging from about 40 to 50 nm for the pristine samples without any heat treatment to about 50 nm and slightly more for the annealed samples. It should also be noted that similar sizes were determined for all the different crystallographic structures; only intermediate phases (quartz-like material) typically show a smaller size. Only a slight increase with increasing heating temperature and annealing time may be stated. However, there are no dramatic changes in the crystallite size, as already found by the analysis of the SEM micrographs (cf. Figure 1). On the other hand, with average values between ca. 40 nm and 50 nm, the crystallite sizes determined by XRD are substantially smaller than those found by SEM investigations, indicating a strong agglomeration of different crystallites.

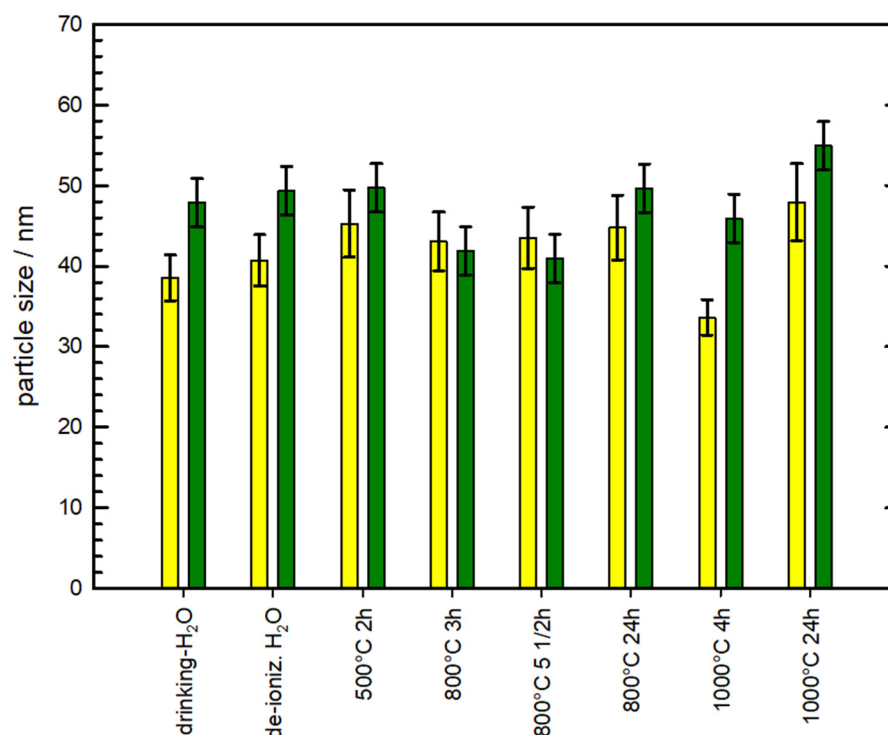


Figure 5. Particle size determined from a quantitative evaluation of the width of the diffraction peaks. Yellow bars correspond to the evaluation of the Scherrer formula, and green bars to the Williamson–Hall analysis, respectively. Error bars indicate the confidence interval of the fits.

3.3. X-ray-Absorption Fine Structure Measurements

A comparison of the Ge K-edge X-ray absorption near edge data (XANES) from Ge-reference compounds and the derived Ge-oxide nanoparticles is presented in Figure 6. As can be seen in Figure 6a, the XANES spectra of the reference materials GeO₂, Ge[OCH(CH₃)₂]₄, and amorphous GeO (a-GeO), a Ge(111) single crystal and an oxidized Ge metal powder are substantially different, in terms of the edge positions, white line features, and post-edge absorption minima and maxima. As can be seen in the inset, the edge position E_0 —as defined by the first maximum of the derivative spectrum—is about 11,103.3 eV for the Ge(111) single crystal sample (purple color code), while it has a value of 11,104.7 eV for amorphous GeO (dark green color, [40]), 11,107.9 eV for β -GeO₂ (red color code), and slightly less for the Ge isopropoxide with a value of 11,106.9 eV (orange color code). For the oxidized Ge powder sample, two edge positions are already visible in the raw data, corresponding to Ge metal (11,102.9 eV) and Ge⁴⁺ oxide (11,108.1 eV), which is well in accordance with recently published XANES data [48]. As can be seen in Figure 6b, the edge positions for the sol–gel derived samples scatter around $11,107.95 \pm 0.1$ eV, suggesting that all samples exhibit a Ge⁴⁺ state, independent from the heat treatment. Looking at the XANES structures, however, small but subtle differences are visible between the different samples, in agreement with the small changes also observed in the XRD data. Very prominent changes in the near-edge X-ray absorption data are thus not expected due to the heat treatments here.

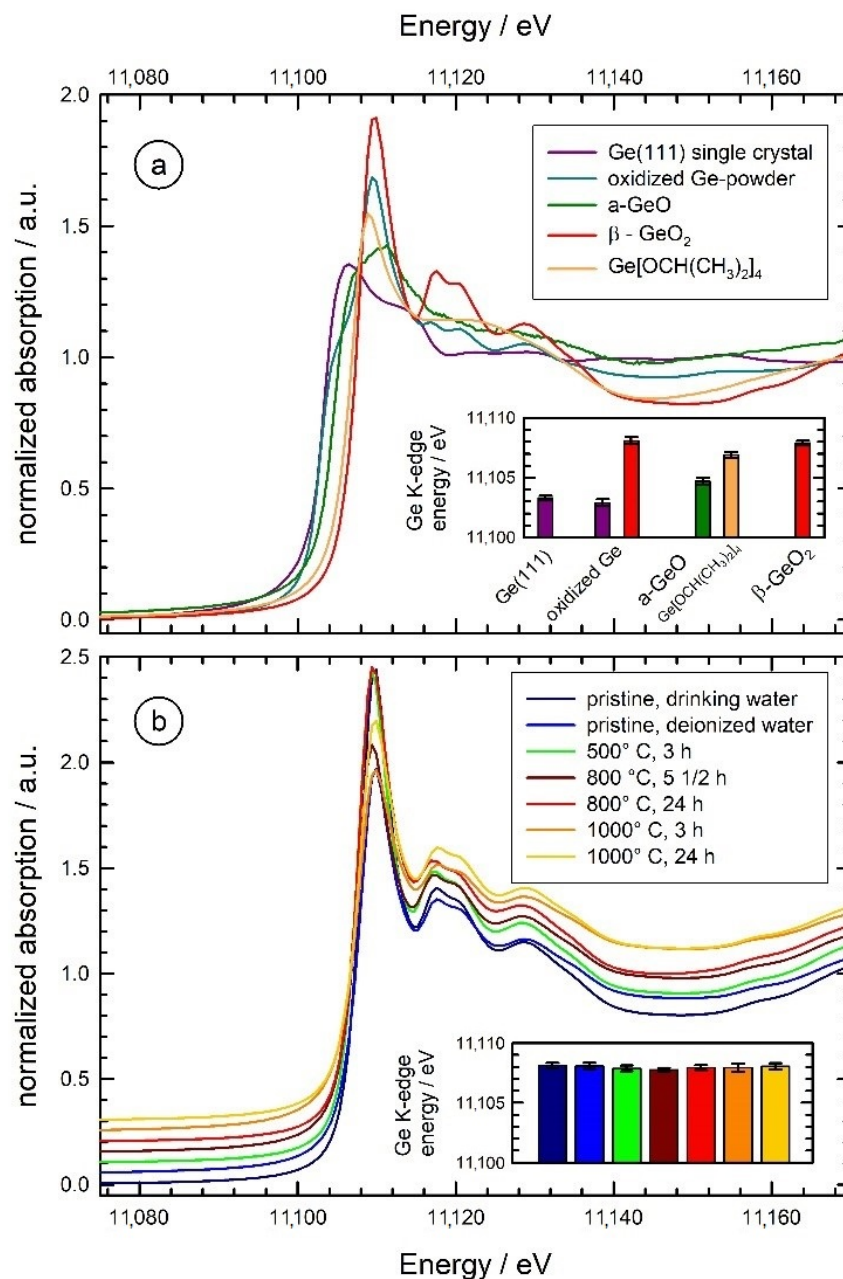


Figure 6. (a) Ge K-edge X-ray absorption near edge data (XANES) of several Ge reference samples as indicated. The inset displays the Ge edge position of those samples. (b) Ge K-edge XANES of several pristine and heat-treated Ge-oxide nanoparticle samples and their edge positions using the same color code as the spectra.

For further data evaluation, the X-ray absorption fine structure $\chi(k)$ was extracted from the background-corrected and normalized experimental absorption data, with the wave number k of the excited photoelectron being equal to $k = (2m(E - E_0))^{1/2}/\hbar$. The k^3 -weighted $\chi(k)$ were Fourier transformed (FT) into R-space, and the magnitude of the $|\text{FT}(\chi(k) \times k^3)|$ as well as the EXAFS fine structures $\chi(k) \times k^3$ of a pristine GeO₂ sample prepared in de-ionized water prior to any heat treatment and a sample after annealing in air for 4 h at 1000 °C are presented in Figure 7. Peaks in the FT correspond to the local atomic environment around the X-ray-absorbing atoms, i.e., the peak at about 1.3 Å can be attributed to the first Ge–O coordination in GeO₂, and the second intense peak at ca. 2.8 Å corresponds to Ge–Ge interactions at larger distances. It has to be stressed here that the peak positions in those Fourier transforms are, in general, not identical with the crystallographic

distances. Because of the photoelectron phase shift arising from the scattering processes, all peaks in the Fourier transform are generally shifted towards lower distances [49]. However, the increased broadening of the Ge-Ge peak in the pristine sample directly after drying indicates a larger degree of disorder in the untreated sample in comparison with the annealed GeO₂ material.

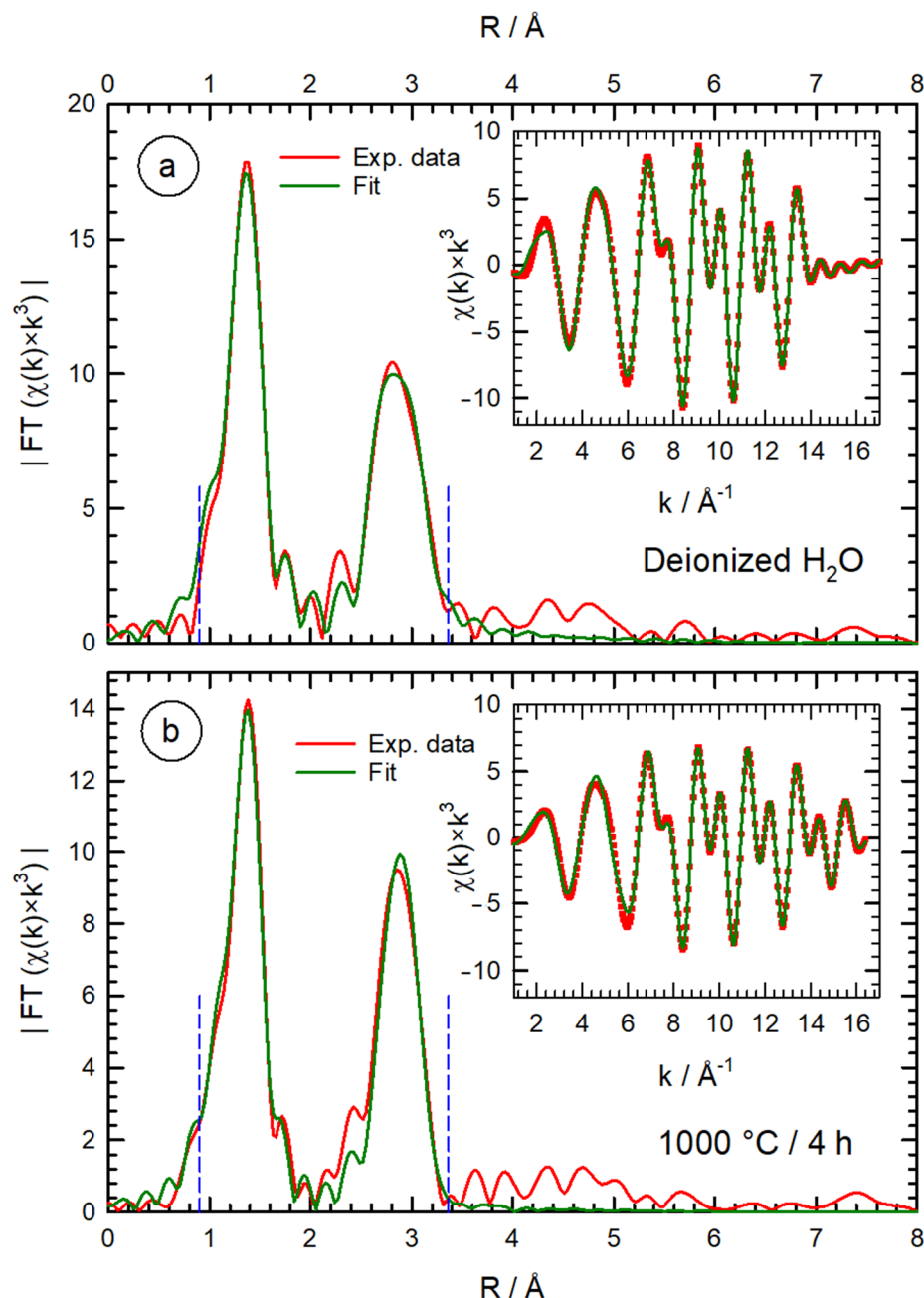


Figure 7. The magnitude of the Fourier transform of the k^3 -weighted EXAFS fine structure $|FT(\chi(k) \times k^3)|$ for (a) the unheated GeO₂ sample prepared in de-ionized water ($R = 0.008$, and (b) after heat treatment in air at 1000 °C for 4 h ($R = 0.016$). Both the raw data and the fits to the model consisting of α - and β -GeO₂ and the quartz-like GeO₂-phase are shown. The insets depict the back-transformed $\chi(k) \times k^3$ data of the Fourier transform in the R -range from $R_{\min} = 0.9 \text{ \AA}$ to $R_{\max} = 3.45 \text{ \AA}$ (dashed vertical blue lines) for the data and the fits. More details are provided in the text (k -range for the Fourier transform: from $k_{\min} = 1.44 \text{ \AA}^{-1}$ to $k_{\max} = 15.95 \text{ \AA}^{-1}$).

Thus, in order to obtain the true distances, as well as the coordination number and the local disorder around the X-ray-absorbing Ge atoms, some fitting of the data is mandatory.

We have, therefore, isolated the structural data related to the first two shells in radial distance from 0.9 Å to 3.4 Å in the FT by means of a filter function, and the data were back-transformed into k-space and fitted with theoretical phase and amplitude functions calculated using FEFF [50]. For the modeling of the EXAFS data, we have employed the crystallographic data of the high and low-temperature phases of α - and β -GeO₂, as well as the quartz-like structure, respectively. According to the results of the XRD studies (see Figures 2–4), the rutile-type GeO₂ was not considered in the EXAFS fits here. For the modeling, all Ge and O atoms up to a distance of 3.5 Å were included in the fit, resulting in three clusters, each of 12 atoms, in total, with nine different single-scattering paths and three additional multiple-scattering paths with significant amplitudes. In principle, each of the considered scattering pathways is related to an individual coordination (i), which itself is characterized by the type of neighboring atom, its distance R_i , the number of atoms in the shell (N_i), the amplitude reduction factor ($S_{0,i}^2$), the inner potential shift ($\Delta E_{0,i}$), and the mean square displacement (σ_i^2). If all involved paths (neighbors) had been treated individually with four parameters per path as detailed above, a huge, statistically insignificant number of fit variables, i.e., $N = 48$ for each GeO₂ species here, would result, and the EXAFS fits would thus not lead to meaningful results if such a multi-parameter model is chosen.

In this context, the limit for the total number of fit variables (N_{idp}) may be calculated from the k-range used for the Fourier transform (i.e., here from $k_{min} = 1.44 \text{ \AA}^{-1}$ to $k_{max} = 15.95 \text{ \AA}^{-1}$), and the range in R used for the fits ($R_{min} = 0.9 \text{ \AA}$ to $R_{max} = 3.45 \text{ \AA}$), for details see, e.g., ref. [51]. For the present experiments, $N_{idp} \approx 23$, so that a suited fit model with an accordingly smaller number of fit parameters is needed here. Thus, we have modeled all the bond distances R_i for the used three structures (i.e., α - and β -GeO₂ and the quartz-like structure) by their effective bond length in the corresponding reference compounds (R_{eff}) multiplied by an expansion factor α_i so that all bond distances of each species are represented by only one single parameter. As the bond distances for the three involved Ge-oxide phases differ substantially, this treatment provides further significance to the fit results. Furthermore, each reference compound is represented by its amplitude reduction factor ($S_{0,i}^2$), its inner potential shift ($\Delta E_{0,i}$), and Debye–Waller factors (σ_i^2) for each type of bond (i.e., Ge–O and Ge–Ge). Thus, only 5 fit parameters are needed for each GeO₂ phase, and in total, 15 fit parameters are necessary for the presented model, substantially less in comparison to N_{idp} . As a consequence, the fits can therefore be assumed to be statistically significant, with fit residuals ($\mathcal{R}$ -factors) in the range of $\mathcal{R} = 0.004$ up to $\mathcal{R} = 0.016$ at maximum, with typical values around 0.008. In Figure 6a,b, representative fits for the sample prepared in de-ionized water (Figure 6a) and the GeO₂ sample annealed at 1000 °C for 4 h (Figure 7b) are presented, respectively. It should be noted that even the fit with the worst $\mathcal{R}$ -factor (Figure 6b) shows an excellent agreement between experimental and fitted data, both in R- and k-space. When the Fourier transformed data from the different samples are compared, it is important to note that small but obvious differences are observed in general, in particular for the peak of the Ge–Ge coordination at about 3 Å radial distance. Accordingly, small differences in the short-range order structure of the various samples are expected. As can be seen, these structural differences in the data were accurately modeled by the fits. Furthermore, it is obvious and well in agreement with the results of the XRD measurements that the amplitudes of the peaks in the EXAFS do not systematically increase with increasing temperature and time, so a substantial growth of the pristine crystallites is unlikely. Instead, the reduced peak intensities in the FT indicate more disorder in the heat-treated powders. For example, contributions from different Ge-oxide species in the samples may qualitatively explain the observed features due to the presence of slightly different bond distances. Furthermore, the presence of 6-coordinated Ge in the argutite GeO₂ species would have resulted in substantially larger nearest neighbor Ge–O peaks in the Fourier transform at small bond distances, which are not observed experimentally. Again, the experimental results of XRD and EXAFS are consistent.

The quantitative fits further suggest that the composition of the prepared GeO_2 materials substantially changes during the heat treatments, in agreement with the XRD results. In particular, the EXAFS fit results suggest slightly larger fractions of quartz-like GeO_2 in the pristine samples prepared in drinking water and de-ionized water without any heat treatment. Also, in the $\beta\text{-GeO}_2$ reference sample, a small contribution of the quartz-like phase was identified (ca. $6.9 \pm 1.2\%$), which may be used as an indicator for the accuracy of all the EXAFS fits, and this error estimate correlates well with the errors determined from the fit uncertainties as given in Figure 8. Similar to the results of the XRD (cf. Figure 3 and Table 1), the high-temperature modification ($\alpha\text{-GeO}_2$) increases from about $19.6 \pm 3.8\%$ after annealing at 500°C for 3 h to about $64.0 \pm 11.2\%$ for 24 h at 1000°C , and the contributions of $\beta\text{-GeO}_2$ and the quartz-like phase decrease from $47.1 \pm 11.6\%$ and $23.8 \pm 2.6\%$ to $8.0 \pm 0.9\%$ and $28.0 \pm 5.0\%$, respectively. In comparison to the results of the XRD, the phase fractions determined by the EXAFS fits are afflicted with larger uncertainties. Measurements of a phase-pure $\alpha\text{-GeO}_2$ sample would have been advantageous in order to further optimize the fit strategy and to foster the results; however, such material was not available for the present investigations. Nevertheless, the trends observed in the quantitative results are very similar to that of the XRD data analysis, and the results of the EXAFS data evaluation are mostly within the confidence intervals of the XRD and vice versa. EXAFS, as well as XRD results, are compiled in Table 1.

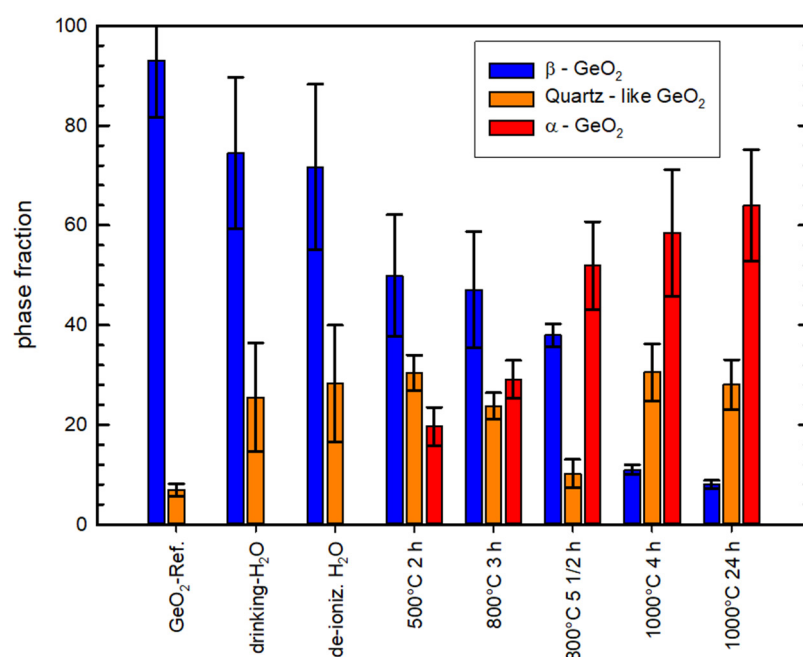


Figure 8. Phase fractions determined by a quantitative analysis of the EXAFS data. The rutile-type argutite GeO_2 structure was not considered in the fits according to the results of the XRD data evaluation.

Looking into some more details about the fit results, one may state that the Ge-O and Ge-Ge distances determined by XRD and EXAFS are very similar, i.e., deviations of the scaling factors α_i from unity used to simultaneously fit all considered scattering contributions in the EXAFS are typically in the order of only ± 0.005 . Thus, the deviation from the crystallographic position of the Ge-O nearest neighbor at about $1.74\text{ \AA}$ results in only $0.02\text{ \AA}$ at maximum, and for the Ge-Ge bonds with a bond distance of about $3.15\text{ \AA}$, the fitted shifts are smaller than about $0.03\text{ \AA}$ at maximum. This implies that the fit results obtained by the EXAFS fits reflect the general accuracy of this X-ray technique [49]. Similarly, the values determined for the thermal and structural disorder (Debye–Waller factors σ_{ji}^2) for the different coordination shells (Ge-O and Ge-Ge) and phases ($\beta\text{-GeO}_2$, quartz-like GeO_2 and $\alpha\text{-GeO}_2$) do not differ substantially for the respective phase in the different

samples, i.e., we have determined a value of $\sigma_{\text{Ge-O}}^2 = 2.73 \times 10^{-3} \text{ \AA}^2 \pm 3.7 \times 10^{-4} \text{ \AA}^2$ for the Ge–O coordination within the β -GeO₂-phase, while values of $\sigma_{\text{Ge-O}}^2 = 2.08 \times 10^{-3} \text{ \AA}^2 \pm 9.0 \times 10^{-4} \text{ \AA}^2$ and $\sigma_{\text{Ge-O}}^2 = 4.43 \times 10^{-3} \text{ \AA}^2 \pm 12.6 \times 10^{-4} \text{ \AA}^2$ were determined for the quartz-like phase and α -GeO₂, respectively. For the Ge–Ge coordination, $\sigma_{\text{Ge-Ge}}^2 = 3.30 \times 10^{-3} \text{ \AA}^2 \pm 3.9 \times 10^{-4} \text{ \AA}^2$, $\sigma_{\text{Ge-Ge}}^2 = 2.43 \times 10^{-3} \text{ \AA}^2 \pm 4.8 \times 10^{-4} \text{ \AA}^2$, and $\sigma_{\text{Ge-Ge}}^2 = 4.83 \times 10^{-3} \pm 11.8 \times 10^{-4} \text{ \AA}^2$ have been determined accordingly. Again, the consistency of those results confirms and justifies the used fit model.

4. Conclusions

In this work, GeO₂ nanoparticles were prepared by hydrolysis of Ge isopropoxide (Ge(OCH(CH₃)₂)₄). The precipitated powders were dried at room temperature and annealed in ambient air using temperatures between 500 °C and 1000 °C for 3 to 24 h. The samples were characterized by a combination of scanning electron microscopy, X-ray diffraction, and X-ray absorption fine structure spectroscopy, revealing the morphology, crystallographic structure, and short-range order structure of the prepared Ge-oxide nanomaterials, respectively. SEM micrographs indicate that the shape of the particles is mainly cubic, with a particle size of 100–150 nm, irrespective of subsequent heat treatment. XRD and EXAFS experiments proved the presence of the low- and the high-temperature GeO₂-phases, i.e., β - and α -GeO₂, and the quartz-like GeO₂ phase. No evidence for other Ge–O phases, such as di-valent GeO, the rutile-type argutite, or amorphous GeO₂, could be found. While the quantitative analysis of XRD patterns alone appears to be difficult due to the similarity of the involved, mainly hexagonal GeO₂ phases, the combination of XRD with X-ray absorption spectroscopy, in particular EXAFS and XANES, provides a substantial significance to the fit results.

According to the quantitative XRD and EXAFS data evaluation, the low-temperature β -GeO₂ phase dominates the pristine samples without any heat treatment, while for intermediate temperatures between 500 °C and 800 °C, substantial contributions of quartz-like GeO₂ were found, with concentrations of ca. 30–50%. With increasing annealing time and temperature, the high-temperature α -GeO₂ was found in increasing amounts, with concentrations of up to 60%. Phase-pure α -GeO₂ nanomaterials could not be synthesized here. Eventually, the annealing temperature, as well as the annealing time, have to be further increased, and the methodology used here should accompany future optimization of the preparation procedures. The rather simple synthesis route used in the present work appears, however, interesting for the preparation of larger amounts of α -GeO₂ nanoparticles and also for a technological application, such as, e.g., the use in Li-ion batteries, as recently reported [52,53], for industrial catalysts [54,55], where, in particular, the high-temperature capabilities and durabilities are important, as well as potentially for pharmaceuticals [56].

In the future, in situ investigations of the Ge-oxide formation processes are planned. As recently shown, a simultaneous combination of quick-scanning EXAFS and X-ray diffraction with a sub-second time resolution allows the following of crystallization processes in the liquid phase in large detail [57,58]. Thus, more information about the dynamics of the extremely fast GeO₂ formation processes may be obtained, and the preparation conditions for a tailored preparation of Ge oxides may be identified. Furthermore, heat treatments can also be followed in situ [59], making more details of the high-temperature α -GeO₂ phase formation accessible. Last but not least, simultaneous and consistent fits of XRD and EXAFS data would be an interesting option.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/cryst13081219/s1>, Figure S1. The structure of rutile-type GeO₂ (spacegroup 136). Each Ge is coordinated with 6 oxygen. The lattice parameters are $a = b = 4.3950 \text{ \AA}$ and $c = 2.8590 \text{ \AA}$, and the lattice is rectangular, i.e., $\alpha = \beta = \gamma = 90^\circ$. Ge is located on position (2a) $x = y = z = 0$, while oxygen is on (4f) with $x = y = 0.3070$ and $z = 0$. The calculated Ge–O distances are $R_1 = 1.866 \text{ \AA}$ and $R_2 = 1.908 \text{ \AA}$. The shortest Ge–Ge distance is $2.859 \text{ \AA}$. Figure S2. The structure of β - (low-temperature) GeO₂ (spacegroup 152). Each Ge is coordinated with 4 oxygen. The lattice parameters are $a = b = 4.9851 \text{ \AA}$ and $c = 5.6455 \text{ \AA}$, and the lattice is hexagonal, i.e., $\alpha = \beta = 90^\circ$

and $\gamma = 120^\circ$. Ge is located on position (3a) $x = 0.4513$, $y = 0$ and $z = 0$, while oxygen is on (6c) with $x = 0.3965$, $y = 0.3022$ and $z = -0.0911$. A shift-vector of $x = y = 0$ and $z = 1/3$ is used. The calculated Ge-O distances are $R_1 = 1.738 \text{ \AA}$ and $R_2 = 1.739 \text{ \AA}$. The shortest Ge-Ge distance is $3.152 \text{ \AA}$. Figure S3. The structure of α - high-temperature GeO_2 (spacegroup 152). Each Ge is coordinated with 4 oxygen. The lattice parameters are $a = b = 4.9851 \text{ \AA}$ and $c = 5.6455 \text{ \AA}$, and the lattice is hexagonal, i.e., $\alpha = \beta = 90^\circ$ and $\gamma = 120^\circ$. Ge is located on position (3a) $x = 0.452$, $y = 0$ and $z = 1/3$, while oxygen is on (6c) with $x = 0.396$, $y = 0.298$ and $z = 0.245$. The calculated Ge-O distances are $R_1 = 1.72 \text{ \AA}$ and $R_2 = 1.76 \text{ \AA}$. The shortest Ge-Ge distance is $3.15 \text{ \AA}$. Figure S4. The structure of the quartz-like GeO_2 structure (spacegroup 154). Each Ge is coordinated with 4 oxygen. The lattice parameters are $a = b = 4.987 \text{ \AA}$ and $c = 5.652 \text{ \AA}$, and the lattice is hexagonal, i.e., $\alpha = \beta = 90^\circ$ and $\gamma = 120^\circ$. Ge is located on position (3a) $x = 0.4513$, $y = 0$ and $z = 0$, while oxygen is on (6c) with $x = 0.3969$, $y = 0.3021$ and $z = 0.0909$. A shift-vector of $x = y = 0$ and $z = 2/3$ is used. The calculated Ge-O distances are $R_1 = 1.737 \text{ \AA}$ and $R_2 = 1.741 \text{ \AA}$. The shortest Ge-Ge distance is $3.152 \text{ \AA}$. Figure S5. (a) Calculated X-ray diffraction patterns for α -, β -, quartz-like and rutile-type GeO_2 and Cu K_α -radiation, respectively. (b) Zoomed view in an angular range with substantial differences between the different GeO_2 -species. Figure S6. (a) Representative Williamson-Hall plots for the GeO_2 -sample annealed at 500°C for 3 h. (a) W-H-plot and size of the low-temperature β - GeO_2 phase. (b) W-H-plot and size of the quartz-like GeO_2 phase. (c) W-H-plot and size of the high-temperature α - GeO_2 phase.

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